

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

17%

global Internet traffic growth

year-over-year

225

major disruptions

observed globally

6.5%

traffic mitigated

security signal



2025 planning lens

Growth, adoption, fragility, and threat signals for technology executives and network infrastructure leaders.

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

YoY global traffic growth

late Nov peak vs start of year

Executive readout

Overall Internet usage expanded steadily through 2024. Peak observed growth reached 17% year-over-year in late November. Starlink growth reflects satellite expansion to underserved locations.

Note: Starlink trend is scaled relative to peak observed request volume and Starlink is not yet globally available.

3.3x

Starlink traffic surge

normalized worldwide request volume



Strategic implication

Capacity planning must account for continued baseline Internet growth plus nonlinear demand from alternative access networks such as satellite broadband.

Google Leads Top 10 Services as Generative AI Rises

Rank	Internet service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Top 10 most popular Internet services worldwide

Generative AI moved into mainstream traffic patterns

ChatGPT led the generative AI services category. Traffic to generative AI services continued its meteoric rise in 2024. Ranking signals demand concentration around platform ecosystems.

SERVICEPLATFORMAI

Leaders should track service concentration, AI workload growth, and provider dependencies as operational and commercial signals.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Chrome drove the inflection; Firefox began rolling out post-quantum by default and Safari began initial testing.

Post-quantum encryption

2% → 12–13%

of TLS 1.3 traffic after Google Chrome 124 enabled post-quantum key agreement by default on April 17.

21%

HTTP/3 adoption

traffic uses QUIC-based HTTP/3

12%

Go API requests

top automated API client language

33%

iOS traffic

mobile OS share

API client language order

Go leads (~12%), followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

major Internet
disruptions

observed globally in 2024

116+

government directed

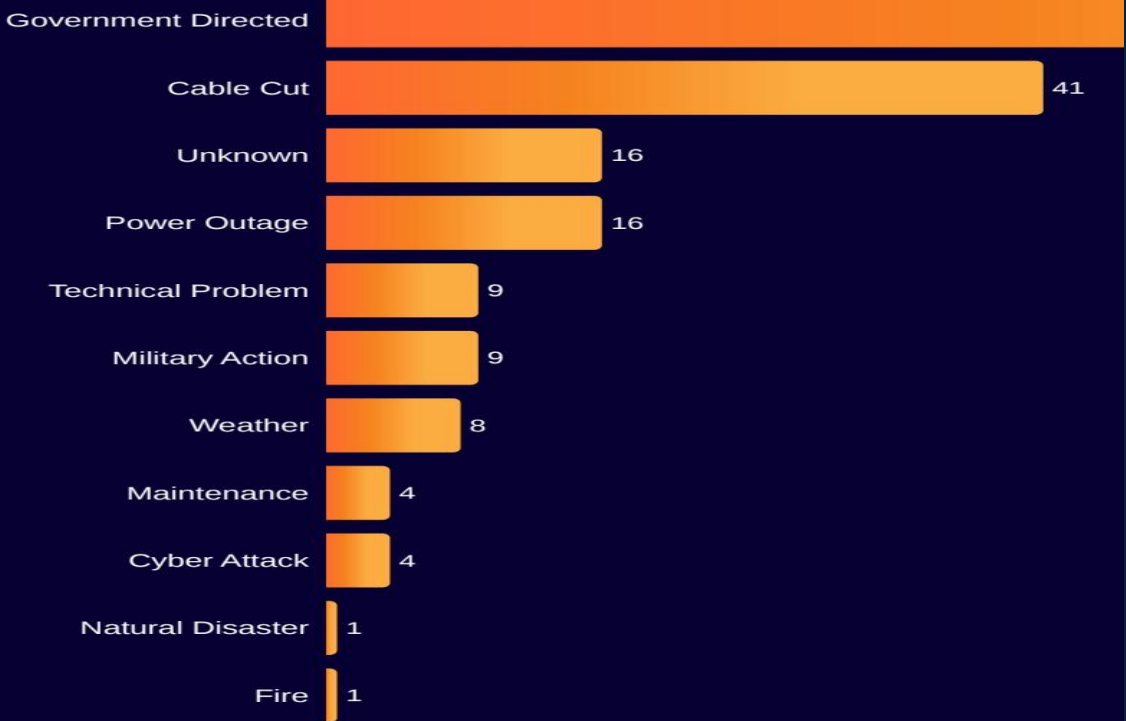
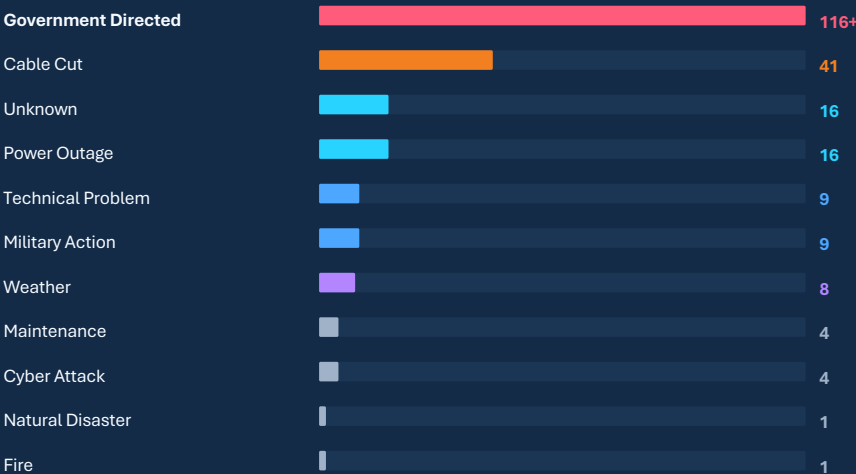
leading outage cause

41

cable cuts

second leading cause

Disruptions by cause



Causes of major Internet disruptions observed in 2024

Resilience planning point

Outage risk is not solely technical: policy actions, physical cable routes, and power dependencies drive material availability exposure.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all processed traffic mitigated

scale of ongoing threats

68%

global bot traffic concentration

originated from top 10 countries

4.3%

emails were malicious

email security analysis

43%

malicious emails with deceptive link

among malicious emails

RPKI

+6.4%

valid routes increased

1

Most attacked sector

Gambling and Games was the most attacked industry category.

2

Log4j persistence

CVE-2021-44228 continued to be actively targeted more than two years after disclosure.

3

Email threat source

.bar originated the largest share of malicious and spam email observed.

Security message: threat management requires layered DDoS/app protection, bot intelligence, patch governance, email controls, and routing security hygiene.

Five Takeaways Shaping the 2025 Internet Outlook

1

Growth baseline rises

Global Internet traffic grew 17% year-over-year; Starlink surged 3.3x as satellite connectivity expanded.

2

Crypto agility accelerates

Post-quantum encryption moved from ~2% to 13% of TLS 1.3 traffic after Chrome 124 default enablement.

3

Availability risk is geopolitical

Government-directed shutdowns led 225 major disruptions, followed by cable cuts and power outages.

4

Threat pressure stays pervasive

6.5% of all traffic was mitigated; Gambling/Games led attacked sectors and Log4j targeting persists.

5

Developer ecosystem shifts

Go became the most popular API client language with roughly 12% of automated API traffic.

2025 outlook

Plan for stronger capacity curves, faster protocol and encryption adoption, higher resilience requirements, and security programs that assume persistent exploit reuse.